



Intestinal microbes derived butyrate is related to the immunomodulatory activities of *Dendrobium officinale* polysaccharide

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ABSTRACT

Although immunomodulatory activities of *Dendrobium officinale* polysaccharide has been investigated for many years, yet the potential contribution of its metabolite derived from intestinal microbes on immunoregulation effect has not been reported. In this study, polysaccharide DOW-5B with average molecular weight of 39.4 kDa was isolated from the stem of *Dendrobium officinale* Kimura et Migo. The carbohydrate content was 91.97% and no protein was detected. The monosaccharide analysis showed this polysaccharide was composed of glucuronic acid and glucose at a molar ratio (M/G) of 1.2:19.4. Animal test indicated DOW-5B increased the diversity of gut microbiota on mice. Beneficial microbes such as *Ruminococcus*, *Eubacterium*, *Clostridium*, *Bifidobacterium*, *Parabacteroides* and *Akkermansia muciniphila* increased while harmful bacteria in *Proteobacteria* decreased. Surprisingly, DOW-5B promoted gut microbes to generate more butyrate and mainly produced by *Parabacteroides*_sp_HGS0025. Further, we found the health of large intestine as well as immunity response of mice was improved. In addition, *Parabacteroides*_sp_HGS0025 positively correlated with butyrate, IgM, IL-10, and TNF- α products in intestine and mice blood, respectively. The data suggested that *Dendrobium officinale* polysaccharide has function on immunity may be mediated by butyrate. It adds new evidence to support the basis of how herbal polysaccharides affect immunity.

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1. Introduction

Dendrobium officinale Kimura et Migo (Tie-Pi-Shi-Hu) distributes in several countries over the world, such as Japan, United States and Australia, and widely spread in China, for example Zhejiang, Anhui, Guangxi, Yunnan and Guizhou. Its stem has been used as a tonic and health food in folk for thousands of years in China and other countries [1,2]. Over the past decades, 190 compounds have been isolated from *Dendrobium officinale* [3]. Since 1994, polysaccharide has been isolated and analyzed [4]. Interestingly, polysaccharides are main component in the dried stem of *Dendrobium officinale*, which has crucial medicinal effects [5]. Bioactivity research on polysaccharide fraction has confirmed that it possess immunoenhancement, anticancer, gastric ulcer protective, hypolipidemic, antihypertensive effects and so on [6]. Among these activities, immunomodulating effects have been paid more attention in recent years. However, most of studies on immunomodulating were performed *in vitro*. Wen-Chuan Lin et al. have reported that oral administration with polysaccharide fraction could enhance immunity *in vivo* [7,8]. Here, it raised a question as to

whether immunomodulation was directly affected by polysaccharide fraction or metabolites produced by gut microbiota. Thus far, it remains unclear.

As we all know, there are a huge number of microbes in our distal gut. The function of these bacteria is a crucial factor for shaping and modulating immune system [9]. Interesting thing is polysaccharides are the major nutrient for shaping gut microbiota [10]. Humans consume dozens of different plant derived glycan in their daily life, while most of them could not digested by enzymes encoded by human genome. However, microbes hosted in our distal of gut could degrade these indigested glycans. *Bacteroidetes*, which occupied around 48% of total bacteria, is one of two dominant phyla in the distal of human gut [11]. *B. thetaiotaomicron* and *B. ovatus* are two species belonging to *Bacteroidetes* phyla and each species has evolved to degrade complex glycan. Taken complex pectin RG-II as an example, *B. thetaiotaomicron* has evolved novel catalytic functions to degrade RG-II into monosaccharide [12]. These data imply that polysaccharide can be metabolized by microbes. Therefore, we hypothesize that herbal polysaccharide may mediate immune response by metabolites produced by gut microbiota.

Short-chain fatty acids (SCFAs), such as acetate, propionate, and butyrate, are the overriding metabolic products by microbiota [13]. They have been appreciated for their beneficial effects, such as energy source and anti-inflammatory on human host, especially butyrate [14,15].

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washed successively with absolute ethanol and acetone for three times. Finally, the precipitation was dried in a drying oven at 50 °C and named as DOW. Then the crude polysaccharide was further fractionated on a DEAE Sepharose Fast Flow column (5 cm × 50 cm) in a gradient way to obtain DOW-5B, which was eluted by 0.05 M NaCl [17].

2.2. Glycosyl composition analysis

2. Experimental

2.1. Isolation of crude polysaccharide DOW-5B

The sugar composition was analyzed by a modified HPLC method described as previously [18]. Briefly, polysaccharide (2 mg) was dissolved in 1 mL H₂O and hydrolyzed with the same volume of 4 M TFA in an ampoule at 110 °C for 4 h, followed to remove TFA by evaporated with MeOH under reduced pressure. The residue was redissolved in 20 µL H₂O, and 50 µL 0.6 M NaOH, 100 µL of PMP (0.5 M methanolic solution) were added into the residue solution. The mixture was kept at 70 °C for 100 min. After cooling to room temperature, the mixture was neutralized with 100 µL of 0.3 M HCl, and extracted with chloroform and water (1:1, v/v) for three times. The aqueous layer was analyzed by HPLC as previously described [19]. The ratio of PMP derivatives is calculated by integration of the corresponding peak area.

2.3. Animals and experimental design

All animal experiments were performed according to guidelines approved by the Institutional Animal Care and Use Committees of Shanghai Institute of Materia Medica, Chinese Academy of Sciences (SCXK-2017-0005).

Twelve 6-week-old C57BL/6 female mice were purchased from Shanghai SLAC Laboratory Animal Center (Shanghai, China). Animals

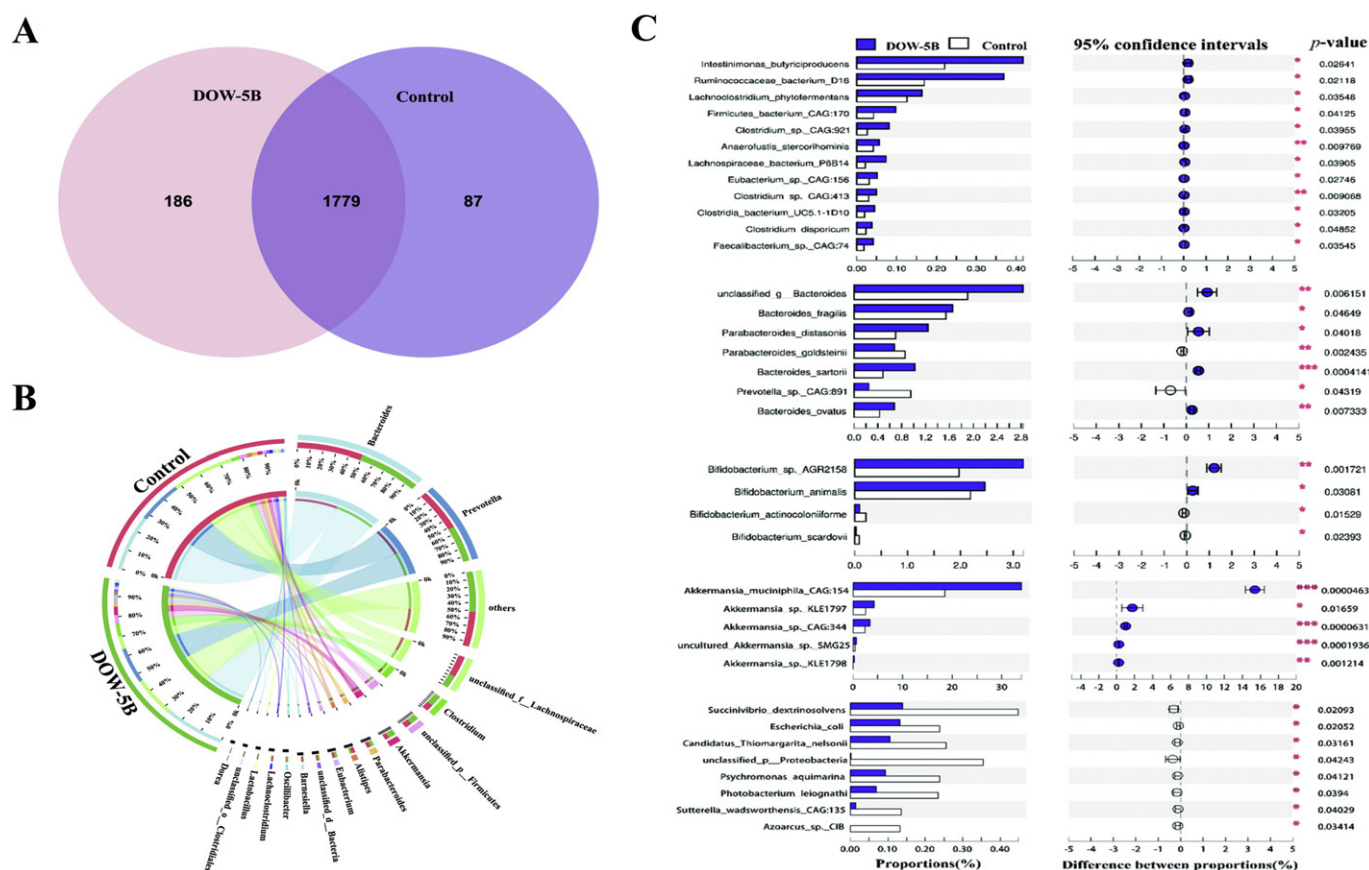


Fig. 1. The composition of microbes in each group and comparison of two groups. (A) The composition of gut microbiota on mice was showed by Venn; (B) the composition of gut microbiota in mice was analyzed from genus level; (C) depend on the five main phyla in the gut, two groups were compared from the species level by Welch's *t*-test.

were housed in two individual cages with free access to food and sterile drinking water (DW) in a temperature-controlled room ($21^{\circ}\text{C} \pm 2^{\circ}\text{C}$) under a 12 hour dark-light cycle. After an accommodation period of 1 week, mice were fed for 25 days with standard chow diet. Mice in DOW-5B group were administrated 0.25% DOW-5B (w/v) in the drinking water, while mice in control group were administrated drinking water freely without any supplementary. Every two days, all mice were weighed and recorded. After 25 days, faeces, whole blood, and colon tissue were collected from each mouse. Faeces were kept in -80°C for SCFAs, faecal lipocalin2 (LCN2) detection and metagenomics analysis. Whole blood was centrifuged to prepare serum for IgA, IgM TNF- α IL-1 β , IL-6 and IL-10 analysis. Based on the weight record, three mice from each group were selected for the above experiments.

2.4. Flow cytometry and ELISA

Mouse IL-1 β , IL-6, TNF α and IL-10 Flex Set kit were purchased from BD™ Cytometric Bead Array. Mouse Lipocalin-2/NGAL kit was purchased from R&D Biosystems. IgA Mouse ELISA Kit and IgM Mouse ELISA Kit were purchased from Abcam. All of the factors were performed according to the manufacturers' instruction.

2.5. Tissue histology

Colon tissues were fixed in 10% neutral buffered-formalin, paraffin embedded, cut into 5- μm sections and H&E stained or Alcian blue stained [20,21]. Muc-2 protein was detected with anti-muc-2 antibody by immunohistochemistry (IHC). Then the expression of muc-2 was quantified by IPP6.0 software.

2.6. Quantification of short-chain fatty acids

Faecal samples were first thawed on wet ice. Then, 1000 μL 0.5% phosphate was added to 100 mg faeces and the samples were thoroughly homogenized by vortex for 2 min. The samples were then centrifuged at 17,949g for 10 min. Subsequently, 800 μL supernatant was pipetted and same amount of ethyl acetate were added. Then, samples were thoroughly homogenized by vortex for 2 min and then centrifuged at 17,949g for 10 min. Finally, 600 μL supernatant were pipetted and 0.5 μM tetramethylpentanoic acid as final concentration were added by vortex. Samples were kept on ice or frozen until quantification of SCFAs by Gas Chromatography–Mass Spectrometry (GC–MS, Agilent 7890A/5975C). An Agilent DB-WAX (30 m $\times$ 0.25 mm ID $\times$ 0.25 μm) was utilized.

2.7. Metagenomic DNA extraction and shotgun metagenomics sequencing

Three samples from each group were selected to prepare faecal DNA. It was extracted using the QIAamp DNA Stool Mini Kit (Qiagen, USA) and fragmented by Covaris M220 following PE library construction (TruSeq™ DNA Sample Prep Kit), bridge PCR (HiSeq 3000/4000 PE Cluster Kit). After above processing, all the samples were sequenced by Illumina HiSeq (HiSeq 3000/4000 SBS Kits).

2.8. Bioinformatics analysis for shotgun metagenomics reads

Adaptor contamination and low quality reads were discarded from the raw sequencing reads, and the remaining reads were filtered to eliminate mice host DNA based on the mice genome reference. All high-quality reads from each sample was assembled to contigs by

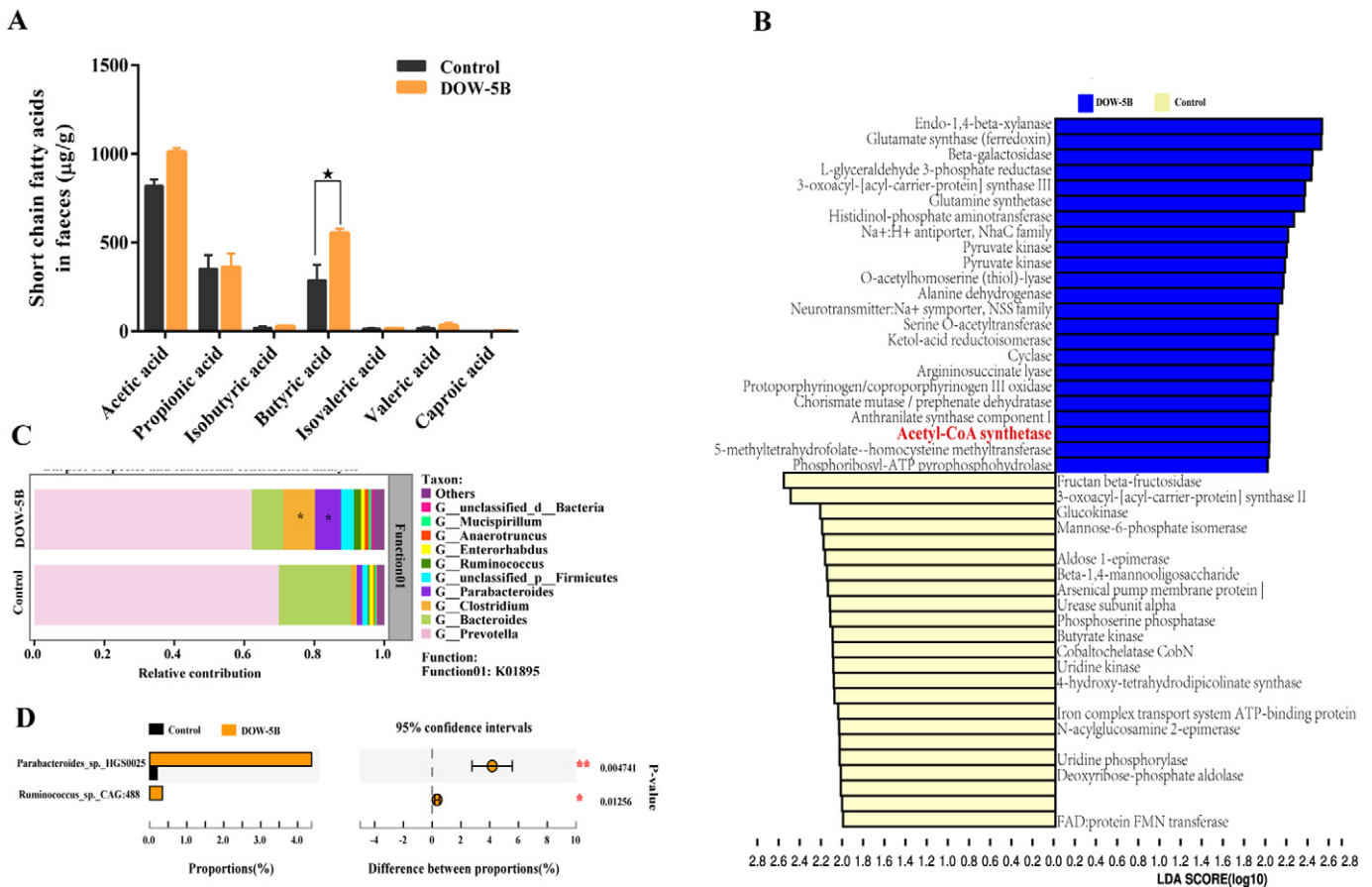


Fig. 2. Gut microbes generated more butyrate and mainly produced by *Parabacteroides_sp_HG-S0025*. (A) SCFAs in the faeces of mice were detected by GC–MS and butyrate increased significantly. (B) LefSe analysis on whole GENE_SET. Acetate CoA-transferase COG0365 significantly enriched in DOW-5B group. (C) Analysis of the contribution between species and function from genus level. (D) Two group were compared based on k01895 gene set by Welch's t-test.

using metaSPAdes from SPAdes v3.10.0 package with default settings, and a nonredundant gene catalog was constructed by MetaGeneMark and CD-HIT. Gene profile was calculated according to the high-speed quantification tool Kallisto 46 and normalized by gene length. Genes were annotated by blasting against the NCBI database using Diamond. The taxonomic composition was acquired by MEGAN and KO composition obtained using the KEGG Orthology-Based Annotation System (KOBAS) tool. Gene counts were calculated by counting the number of genes in each sample. Welch's *t*-test was used in STAMP analysis. All *p*-values were two-sided and were considered as statistical significance below 0.05. Finally, all the data were analyzed on the free online platform of Majorbio I-Sanger Cloud Platform (www.i-sanger.com).

3. Results and discussions

3.1. Monosaccharide composition

Sugar composition analysis by HPLC based on precolumn derivatization with PMP indicated that DOW-5B contains glucuronic acid and glucose at a molar ratio (M/G) of 1.2:19.4 (Fig. S1). The molecular weight of DOW-5B is 39.4 kDa (Fig. S2).

3.2. DOW-5B raised the beneficial bacteria while decreased harmful one in the gut of mice

After 25 days treatment by DOW-5B, there was no significant change on the weight of mice (Fig. S3). Glycan is one of the major factors to shape the gut microbiota [10]. Hence, for clarifying the linking between mice and DOW-5B, gut bacteria were analyzed by metagenomics sequencing. It was evident that the diversity of gut bacteria was more enrichment after DOW-5B treatment (Fig. 1A; Table S1). Especially, the level of butyrate-producing *Clostridium*, *Parabacteroides*, and anti-

inflammatory *Akkermansia* were higher than control group (Fig. 1B; Table S2). Following, five main phyla including *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, *Verrucomicrobia*, and *Proteobacteria* were analyzed by Welch's *t*-test on species level based on RPKM (Fig. 1C; Table S3) [22].

For *Firmicutes* phyla, including *Intestinimonas butyriciproducens*, *Ruminococcaceae bacterium*, *Clostridium* and *Eubacterium* are butyrate generation bacteria [23,24]. In this study, all of these bacteria increased significantly ($p < 0.05$). *B. fragilis*, *teroides distasonis* and *B. ovatus*, which belong to *Bacteroides*, were significantly increased ($p < 0.05$). Interestingly, these bacteria have been reported in regulating immunity, generating butyrate and glycan degrading [16,25,26]. For *Actinobacteria* phyla, *B. animal* has been applied as probiotics such as BB12 for years [27]. It was significantly different compared with control group ($p < 0.05$) in this study. *Akkermansia muciniphila*, which belongs to *Verrucobacteria*, is a newly described mucin-degrading species improved metabolic disorder disease [28]. In this research, we found *Akkermansia muciniphila* CAG:154 increased significantly after DOW-5B treatment ($p < 0.05$). *Proteobacteria* phyla usually relate with most of pathogenicity strains. For example, *E. coli* is one of the members in this phyla. After analysis the *Proteobacteria* phyla, we were surprised to find species in this phyla decreased significantly after DOW-5B treatment ($p < 0.05$). Here we should ask one question what is the function of the improved microbes. We know SCFAs are the main metabolites produced by gut microbes [14]. In addition, there are a growing data reported the function of SCFAs on immunoregulation [28]. Hence, we next asked whether gut microbes could generate SCFAs after DOW-5B treatment.

3.3. Butyrate significantly increased in the faeces of mice and mainly generated by *Parabacteroides* sp_HGS0025

SCFAs are key bacterial metabolites, in particularly acetate, butyrate, and propionate [13]. Interestingly, butyrate induces the differentiation

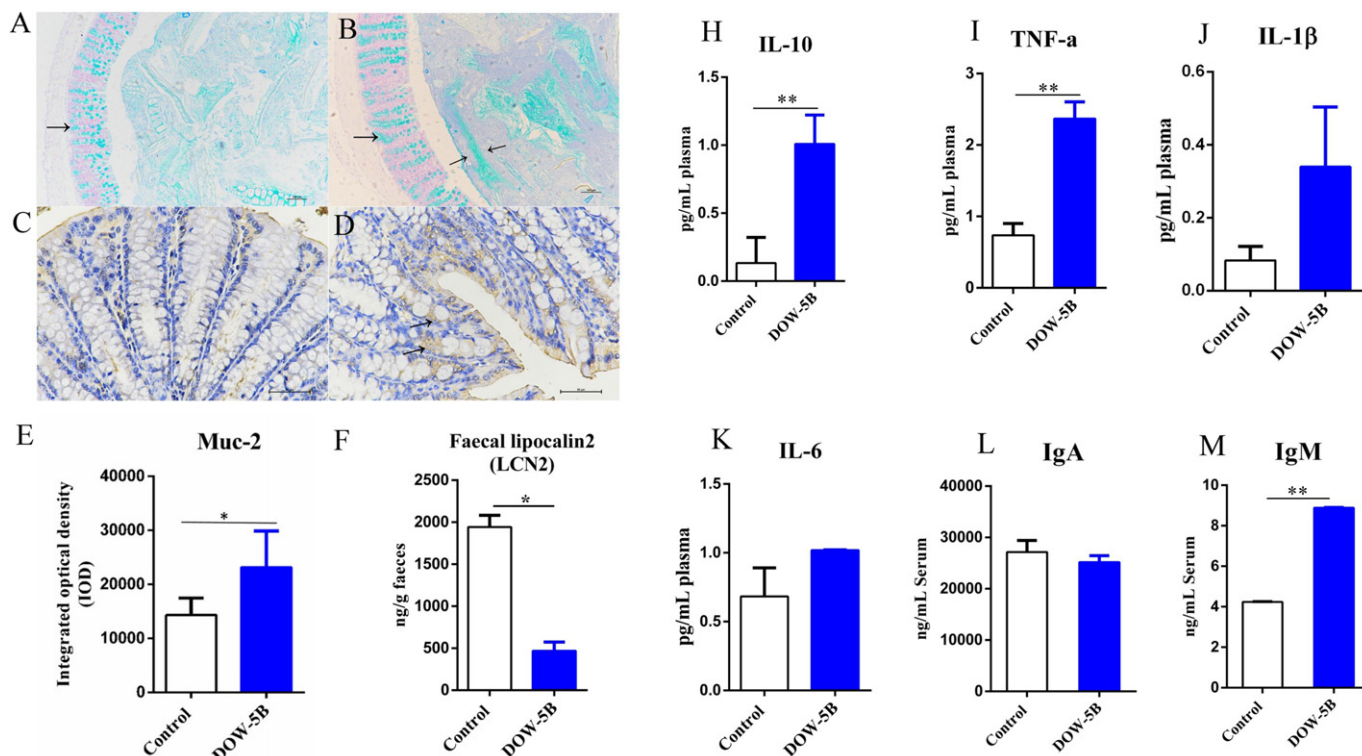


Fig. 3. Polysaccharide DOW-5B improved the immunity on mice. (A) Alcian blue-stained colonic sections showing the mucus (arrow) on control mice; (B) Alcian blue-stained colonic sections showing the mucus and mucus layer (arrow) on DOW-5B treated mice; (C) the expression of mucin-2 on colon by IHC staining in the control group; (D) the expression of mucin-2 on colon by IHC staining (arrows) in the DOW-5B group; (E) quantified mucin-2 by IPP6.0 software. (F) Faecal lipocalin2 was detected by ELISA ($n = 3$). (H) IL-10 was tested by flow cytometer ($n = 3$); (I) TNF- α was tested by flow cytometer ($n = 3$); (J) IL-1 β was tested by flow cytometer ($n = 3$); (K) IL-6 was tested by flow cytometer ($n = 3$); (L) IgA was tested by ELISA ($n = 3$); (M) IgM was tested by ELISA ($n = 3$). * represented $p < 0.05$; ** represented $p < 0.01$.

of colonic regulatory T cells [24,29]. In our study, the SCFAs from faeces of mice were detected by GC–MS. As we expected, Mice treated with DOW-5B generated more butyrate ($p < 0.05$) and acetic acids ($p > 0.05$) than control group (Fig. 2A). This result drives us to investigate which bacteria generate SCFAs after DOW-5B treatment.

For answer which bacteria are the main butyrate-producer, we should know which synthetase is responsible for the butyrate synthesis. Previously, two different pathways were identified for butyrate production in microbe [30,31]. In these two pathways, butyryl-CoA and butyrate kinase are two key genes for each pathway. Butyryl-CoA was found in majority of human microbe while butyrate kinase was only identified in a few bacteria. In this study, four dominant phyla including *Bacteroides*, *Firmicutes*, *Verrucomicrobia*, and *Proteobacteria* were analyzed by LEfSe. To our surprised, butyryl-CoA:acetate CoA-transferase COG0365 significantly enriched in DOW-5B group (Fig. 2B; Table S4). Following, a gene set k01895, which is butyryl-CoA, was constructed. Based on this gene set, two groups were analyzed. Results showed that *Parabacteroides* and *Clostridium* have more contribution on k01895 gene compared with control group (Fig. 2C; Table S5) on genus level. *Parabacteroides*_sp._HGS0025 and *Ruminococcus* which belong to *Bacteroidetes* and *Firmicutes* phyla, respectively, were

significantly different from control group ($p < 0.01$, $p < 0.05$) (Fig. 2D; Table S6). Interestingly, both *Parabacteroides* and *Ruminococcus* species have been reported to generate butyrate [11,16]. Till here, we can clearly understand *Parabacteroides* and *Ruminococcus* species are the key bacteria to generate butyrate under DOW-5B treatment. Next, we would like to know the function of the metabolite butyrate derived from DOW-5B on mice.

3.4. Mucus barrier of colon and immunoregulation on mice were improved

Mucin 2 (muc-2) is particularly prominent in the gut where it is secreted from goblet cells in the epithelial lining into the lumen of the large intestine [32]. This is an insoluble mucous barrier serves to protect the intestinal epithelium. In this research, compared with control mice, there was no pathological change after DOW-5B treatment by HE-staining (Fig. S4). Alcian blue-stained colonic sections showed that there was more mucus after DOW-5B treatment (Fig. 3A and B). Moreover, we found muc-2 less expressed in the control group, while high expressed in colon after DOW-5B treatment (Fig. 3C and D). Quantified muc-2 by IPP6.0 software, muc-2 protein was significantly different from control group (Fig. 3E). In addition, LCN2, which is an intestinal

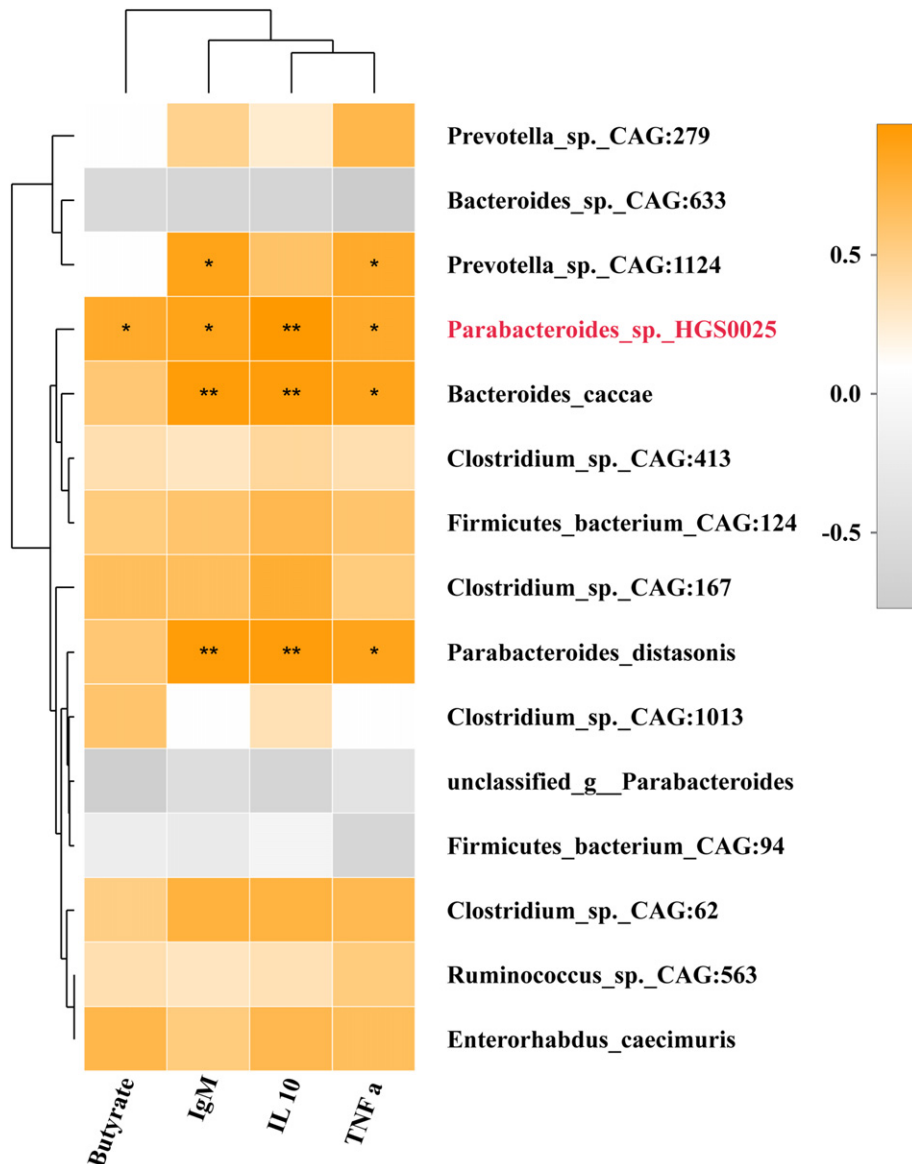


Fig. 4. Analysis the correlation between the species and immune factors and SCFAs by heat map.

inflammation marker, decreased significantly after DOW-5B treatment (Fig. 3F). These results suggest that mucus barrier of colon was improved. It also implied that DOW-5B facilitated the health of large intestine on mice. Following, IL-1 β , IL-6, TNF- α , and IL-10 in serum were tested by Flow cytometry. IgA and IgM in serum were measured by ELISA, respectively. Immune factors IL-10 and TNF- α were significantly different compared with control group (Fig. 3H and I), while IL-1 β , IL-6 and IgA were not significantly different compared with control group (Fig. 3J, K and L). Interestingly, IgM increased significantly in the serum after DOW-5B treatment (Fig. 3M). Our results demonstrated that the immune response of mice was improved after DOW-5B treatment.

3.5. *Parabacteroides_sp_HGS0025* positively co-related with both immune factors and SCFAs

Since *Parabacteroides* and *Ruminococcus* species increased in the DOW-5B group, we want to know the relationship of this species with immune factors, and SCFAs. Thus, co-relation analysis was applied to the gene set k01895. *Bacteroides* species including *Bacteroides caccae*, *Bacteroides_sp_3_1_13*, *Parabacteroides distasonis*, and *Parabacteroides_sp_HGS0025* positively co-related with IgM, IL-10, and TNF- α (Fig. 4; Table S7). In addition, *Parabacteroides_sp_HGS0025* positively co-related with butyrate (Fig. 4; Table S7). From this result, we knew *Parabacteroides_sp_HGS0025* co-related with most of the factors we studied. It suggested that *Parabacteroides_sp_HGS0025* might be one major species involved in the butyrate-producer after DOW-5B treatment.

4. Conclusion

In summary, DOW-5B with average molecular weight 39.4 kDa was isolated from the stem of *Dendrobium officinale* Kimura et Migo. It was composed of glucuronic acid and glucose at a molar ratio (M/G) of 1.2:19.4. Bioactivity test showed this polysaccharide could increase the diversity of gut microbiota on mice. Especially the level of butyrate-producing *Ruminococcus*, *Eubacterium*, *Clostridium*, *Parabacteroides* and anti-inflammatory *Akkermansia* were higher than control mice. To our surprised, butyrate significantly increased and mainly generated by *Parabacteroides_sp_HGS0025*. At the same time, the health of large intestine as well as immunity response of mice was improved. In addition, *Parabacteroides_sp_HGS0025* positively co-related with butyrate, IgM, IL-10, and TNF- α . Our results uncovered polysaccharide isolated from *Dendrobium officinale* might regulate the immunity by SCFAs. These data imply that we may need to explore the function of herbal polysaccharides from their metabolites in the gut.

CRediT authorship contribution statement

Meixia Li: Investigation, Formal analysis. **Han Yue:** Investigation. **Ye-Qing Wang:** Investigation. **Ci-Liang Guo:** Investigation. **Zhenyun Du:** Investigation. **Can Jin:** Investigation. **Kan Ding:** Methodology, Formal analysis.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2020.01.305>.

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